```
from django.shortcuts import render
from django.http import HttpResponse
from django.http import JsonResponse
from .models import mode_selected
from .models import devices
from .models import sensors
from .models import threshold
from .models import mode1
from .models import mode2
from .models import mode3
from .models import mode4
from .models import mode1_vision_system
from .models import mode2_vision_system
from .models import mode3_vision_system
from .models import mode4_vision_system

from datetime import datetime
from datetime import date
from time import sleep

import sys
import numpy as np
import cv2
import re
from plantcv import plantcv as pcv

def mainPage(response):

    mode_selected_obj_global = mode_selected.objects.latest('date')
    devices_obj_global = devices.objects.latest('date')

    mode1_obj_global = mode1.objects.latest('date')
    mode2_obj_global = mode2.objects.latest('date')
    mode3_obj_global = mode3.objects.latest('date')
    mode4_obj_global = mode4.objects.latest('date')

    # Create instances so you can insert into the database
    mode_selected_ = mode_selected()
    devices_ = devices()
    devices_2 = devices()
    sensors_ = sensors()
    threshold_ = threshold()

    mode1_ = mode1()
    mode2_ = mode2()
    mode3_ = mode3()
    mode4_ = mode4()

    mode1_vision_system_ = mode1_vision_system()
    mode2_vision_system_ = mode2_vision_system()
    mode3_vision_system_ = mode3_vision_system()
    mode4_vision_system_ = mode4_vision_system()

    if response.POST.get('action') == 'setup':

        print(" ")
        print("--------------------------- GrowSmart Initializing ----------------------------")
```

```
    print("Time: " + datetime.now().strftime('%Y-%m-%d %H:%M:%S'))
    print("Mode: " + str(mode_selected_obj_global.modeNumber))
    print("Grid: " + mode_selected_obj_global.grid)
    print(" ")
    print(" ")

    json = {
    'modeNumber' :mode_selected_obj_global.modeNumber
    }
    return JsonResponse(json)

if response.POST.get('action') == 'getSensorValues':

    threshold_obj_global = threshold.objects.latest('date')

    print(" ")
    print("------------------------- Checking Sensor Readings -------------------------")
    print("Time: " + datetime.now().strftime('%Y-%m-%d %H:%M:%S'))

    currentMoisture = 34
    temperature = 25
    temperature2 =24
    humidity = 68
    humidity2 = 70

    averageTemperature = (temperature + temperature2) / 2
    averageHumidity = (humidity + humidity2) / 2

    temperatureStatus = 'good'
    humidityStatus = 'good'
    soilMoistureStatus = 'good'

    temperatureStatusSummary = "Default"
    humidityStatusSummary = "Default"
    soilMoistureStatusSummary = "Default"

    if(averageTemperature <= threshold_obj_global.temperature_low):
        temperatureStatus = 'low' # Too Low
    elif(averageTemperature >= threshold_obj_global.temperature_high):
        temperatureStatus = 'high' # Too High
    else:
        temperatureStatus = 'good' # Good

    if (averageHumidity <= threshold_obj_global.humidity_low):
        humidityStatus = 'low' # Too Low
    elif (averageHumidity >= threshold_obj_global.humidity_high):
        humidityStatus = 'high' # Too High
    else:
        humidityStatus = 'good' # Good

    if (currentMoisture <= threshold_obj_global.moisture_dry):
        soilMoistureStatus = 'dry'; # Dry
    elif (currentMoisture >= threshold_obj_global.moisture_dry and currentMoisture <= threshold_obj_global.moisture_wet):
        soilMoistureStatus = 'moist'; # Moist
    elif (currentMoisture >= threshold_obj_global.moisture_wet):
        soilMoistureStatus = 'wet'; # Wet
```

```
if(temperatureStatus == 'high'):
    temperatureStatusSummary = 'Too High!'
elif (temperatureStatus == 'low'):
    temperatureStatusSummary = 'Too Low!'
else:
    temperatureStatusSummary = 'Good'

if (humidityStatus == 'high'):
    humidityStatusSummary = 'Too High!'
elif (humidityStatus == 'low'):
    humidityStatusSummary = 'Too Low!'
else:
    humidityStatusSummary = 'Good'

print("Soil Moisture: "+ str(currentMoisture))
print("Temperature Left: " + str(temperature))
print("Temperature Right: "+ str(temperature2))
print("Average Temperature: "+ str(round(averageTemperature, 2)))
print("Humidity Left: "+ str(humidity))
print("Humidity Rght: "+ str(humidity2))
print("Average Humidity: "+ str(round(averageHumidity, 0)))

# Adaptive Irrigation System Code
if (soilMoistureStatus == 'dry'):
    soilMoistureStatus = 'Dry!'
    print(" ")
    print("~ Soil moisture is dry, adaptive irrigation system activated~")
    print(" ")
    devices_.fansStatus = devices_obj_global.fansStatus
    devices_.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = 'On'
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()

    sleep(1)

    print(" ")
    print("~Adaptive irrigation system deactivated~")
    print(" ")
    devices_2.fansStatus = devices_obj_global.fansStatus
    devices_2.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_2.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_2.calibrationStatus = devices_obj_global.calibrationStatus
    devices_2.waterStatus = 'Off'
    devices_2.seedStatus = devices_obj_global.seedStatus
    devices_2.save()

elif (soilMoistureStatus == 'moist'):
    soilMoistureStatus = 'Moist'
elif (soilMoistureStatus == 'wet'):
    soilMoistureStatus = 'Wet!'

if(temperatureStatus == 'low'):
    print(" ")
    print("~Temperatures is low, fans deactivated~")
```

```
    print(" ")
    devices_.fansStatus = 'Off'
    devices_.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()
elif(temperatureStatus == 'high'):
    print(" ")
    print("~Temperatures is high, fans activated~")
    print(" ")
    devices_.fansStatus = 'On'
    devices_.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()
elif(temperatureStatus != 'low' and humidityStatus == 'high'):
    print(" ")
    print("~Temperatures is not low and humidity is high, fans activated~")
    print(" ")
    devices_.fansStatus = 'On'
    devices_.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()

sensors_.temperature = round(averageTemperature, 2)
sensors_.humidity = round(averageHumidity, 0)
sensors_.moisture = currentMoisture
sensors_.temperatureStatus = temperatureStatusSummary
sensors_.humidityStatus = humidityStatusSummary
sensors_.soilMoistureStatus = soilMoistureStatus
sensors_.save()

sensors_obj = sensors.objects.latest('date')
mode_selected_obj_first = mode_selected.objects.first()
mode_selected_obj = mode_selected.objects.latest('date')

date1 = mode_selected_obj_first.date
date2 = sensors_obj.date

def numOfDays(date1, date2):
    return (date2-date1).days

mode_selected_.daysCounter = numOfDays(date1, date2) + 1
mode_selected_.grid = mode_selected_obj.grid
mode_selected_.rows = mode_selected_obj.rows
mode_selected_.columns = mode_selected_obj.columns
mode_selected_.modeNumber = mode_selected_obj.modeNumber
mode_selected_.image = mode_selected_obj.image
mode_selected_.save()
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        mode_selected_obj_2 = mode_selected.objects.latest('date')

        print("Day "+ str(mode_selected_obj_2.daysCounter))
        print(" ")
        print(" ")

        json = {
        'daysCounter_json' : str(mode_selected_obj_2.daysCounter),
        'date_json': str(datetime.now().strftime('%b. %d, %Y, %-I:%M %p')),
        'temperature_json': sensors_obj.temperature,
        'humidity_json': sensors_obj.humidity,
        'soilMoisture_json': sensors_obj.moisture,
        'temperatureStatus_json' : sensors_obj.temperatureStatus,
        'humidityStatus_json' : sensors_obj.humidityStatus,
        'soilMoistureStatus_json' : sensors_obj.soilMoistureStatus,
        }

        return JsonResponse(json)

    if response.POST.get('action') == 'snapImage':
        mode_selected_obj = mode_selected.objects.latest('date')
        if(mode_selected_obj.modeNumber == 1):
            print(" ")
            print("~[ Mode 1 ] Vision System Starting~")
            print(" ")
            print(" ")

            getTime = datetime.now().strftime('%Y-%m-%d-%H:%M:%S')

            class options:
                def __init__(self):
                    self.debug = "plot"
                    self.outdir = "./assets/gardenPics/"


            args = options()
            #pcv.params.debug = args.debug

            plant_area_list = [] #Plant area array for storage

            #img, path, filename = pcv.readimage(filename='./assets/gardenPics/' + getTime + '.jpg', modeNumber="native") # Read
image to be used
            img, path, filename = pcv.readimage(filename= './assets/gardenPics/test.jpg', mode="native") # Read image to be used

            # START of  Multi Plant Workflow https://plantcv.readthedocs.io/en/stable/multi-plant_tutorial/

            # STEP 1: Check if this is a night image
            # STEP 2: Normalize the white color so you can later
            img1 = pcv.white_balance(img, roi = (600,70,20,20))
            # STEP 3: Rotate the image so that plants line up with grid
            # STEP 4: Shift image
            # STEP 5: Convert image from RGB colorspace to LAB colorspace Keep only the green-magenta channel (grayscale)
            a = pcv.rgb2gray_lab(rgb_img=img1, channel='a')
            # STEP 6: Set a binary threshold on the saturation channel image
            img_binary = pcv.threshold.binary(gray_img=a, threshold=119, max_value=255, object_type='dark')
            # STEP 7: Fill in small objects (speckles)
            fill_image = pcv.fill(bin_img=img_binary, size=100)
```

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# STEP 8: Dilate so that you don't lose leaves (just in case)
dilated = pcv.dilate(gray_img=fill_image, ksize=2, i=1)
# STEP 9: Find objects (contours: black-white boundaries)
id_objects, obj_hierarchy = pcv.find_objects(img=img1, mask=dilated)
# STEP 10: Define region of interest (ROI)
roi_contour, roi_hierarchy = pcv.roi.rectangle(img=img1, x=100, y=160, h=390, w=780)
# STEP 11: Keep objects that overlap with the ROI
roi_objects, roi_obj_hierarchy, kept_mask, obj_area = pcv.roi_objects(img=img1, roi_contour=roi_contour,
                                                      roi_hierarchy=roi_hierarchy,
                                                      object_contour=id_objects,
                                                      obj_hierarchy=obj_hierarchy,
                                                      roi_type='partial')

# END of Multi Plant Workflow

# START of Create Multiple Regions of Interest (ROI) https://plantcv.readthedocs.io/en/stable/roi_multi/

# Make a grid of ROIs
roi1, roi_hier1  = pcv.roi.multi(img=img1, coord=(180,260), radius=50, spacing=(150, 200), nrows=2, ncols=5)


# Loop through and filter each plant, record the area
for i in range(0, len(roi1)):
    roi = roi1[i]
    hierarchy = roi_hier1[i]
    # Find objects
    filtered_contours, filtered_hierarchy, filtered_mask, filtered_area = pcv.roi_objects(
        img=img, roi_type="partial", roi_contour=roi, roi_hierarchy=hierarchy, object_contour=roi_objects,
        obj_hierarchy=roi_obj_hierarchy)

    # Record the area
    plant_area_list.append(filtered_area)

    if(i<10):
        print(plant_area_list[i])

# END of Create Multiple Regions of Interest (ROI)

# Label area by plant ID, leftmost plant has id=0
plant_area_labels = [i for i in range(0, len(plant_area_list))]

#out = args.outdir
# Create a new measurement
pcv.outputs.add_observation(variable='plant_area', trait='plant area ',
                            method='plantcv.plantcv.roi_objects', scale='pixels', datatype=list,
                            value=plant_area_list, label=plant_area_labels)

# Print areas to XML
#pcv.print_results(filename="./assets/gardenPics/plant_area_results.xml")


mode1_vision_system_.image = '../assets/gardenPics/' + 'test' + '.jpg'
mode1_vision_system_.plant1 = plant_area_list[0]
mode1_vision_system_.plant2 = plant_area_list[1]
mode1_vision_system_.plant3 = plant_area_list[2]
mode1_vision_system_.plant4 = plant_area_list[3]
mode1_vision_system_.plant5 = plant_area_list[4]
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mode1_vision_system_.plant6 = plant_area_list[5]
mode1_vision_system_.plant7 = plant_area_list[6]
mode1_vision_system_.plant8 = plant_area_list[7]
mode1_vision_system_.plant9 = plant_area_list[8]
mode1_vision_system_.plant10 = plant_area_list[9]

if(plant_area_list[0] < 721):
    status1 = 'Early'
    image1 = '../assets/icons/seedIcon.png'
elif((plant_area_list[0] > 720) and (plant_area_list[0] < 861)):
    status1 = 'Dvlping'
    image1 = '../assets/icons/plantIcon.png'
else:
    status1 = 'Mature'
    image1 = '../assets/icons/pechayIcon.png'

if(plant_area_list[1] < 721):
    status2 = 'Early'
    image2 = '../assets/icons/seedIcon.png'
elif((plant_area_list[1] > 720) and (plant_area_list[1] < 861)):
    status2 = 'Dvlping'
    image2 = '../assets/icons/plantIcon.png'
else:
    status2 = 'Mature'
    image2 = '../assets/icons/pechayIcon.png'

if(plant_area_list[2] < 721):
    status3 = 'Early'
    image3 = '../assets/icons/seedIcon.png'
elif((plant_area_list[2] > 720) and (plant_area_list[2] < 861)):
    status3 = 'Dvlping'
    image3 = '../assets/icons/plantIcon.png'
else:
    status3 = 'Mature'
    image3 = '../assets/icons/pechayIcon.png'

if(plant_area_list[3] < 721):
    status4 = 'Early'
    image4 = '../assets/icons/seedIcon.png'
elif((plant_area_list[3] > 720) and (plant_area_list[3] < 861)):
    status4 = 'Dvlping'
    image4 = '../assets/icons/plantIcon.png'
else:
    status4 = 'Mature'
    image4 = '../assets/icons/pechayIcon.png'

if(plant_area_list[4] < 721):
    status5 = 'Early'
    image5 = '../assets/icons/seedIcon.png'
elif((plant_area_list[4] > 720) and (plant_area_list[0] < 861)):
    status5 = 'Dvlping'
    image5 = '../assets/icons/plantIcon.png'
else:
    status5 = 'Mature'
    image5 = '../assets/icons/pechayIcon.png'

if(plant_area_list[5] < 721):
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```
    status6 = 'Early'
    image6 = '../assets/icons/seedIcon.png'
elif((plant_area_list[5] > 720) and (plant_area_list[5] < 861)):
    status6 = 'Dvlping'
    image6 = '../assets/icons/plantIcon.png'
else:
    status6 = 'Mature'
    image6 = '../assets/icons/pechayIcon.png'

if(plant_area_list[6] < 721):
    status7 = 'Early'
    image7 = '../assets/icons/seedIcon.png'
elif((plant_area_list[6] > 720) and (plant_area_list[6] < 861)):
    status7 = 'Dvlping'
    image7 = '../assets/icons/plantIcon.png'
else:
    status7 = 'Mature'
    image7 = '../assets/icons/pechayIcon.png'

if(plant_area_list[7] < 721):
    status8 = 'Early'
    image8 = '../assets/icons/seedIcon.png'
elif((plant_area_list[7] > 720) and (plant_area_list[7] < 861)):
    status8 = 'Dvlping'
    image8 = '../assets/icons/plantIcon.png'
else:
    status8 = 'Mature'
    image8 = '../assets/icons/pechayIcon.png'

if(plant_area_list[8] < 721):
    status9 = 'Early'
    image9 = '../assets/icons/seedIcon.png'
elif((plant_area_list[8] > 720) and (plant_area_list[8] < 861)):
    status9 = 'Dvlping'
    image9 = '../assets/icons/plantIcon.png'
else:
    status9 = 'Mature'
    image9 = '../assets/icons/pechayIcon.png'

if(plant_area_list[9] < 721):
    status10 = 'Early'
    image10 = '../assets/icons/seedIcon.png'
elif((plant_area_list[9] > 720) and (plant_area_list[9] < 861)):
    status10 = 'Dvlping'
    image10 = '../assets/icons/plantIcon.png'
else:
    status10 = 'Mature'
    image10 = '../assets/icons/pechayIcon.png'

mode1_vision_system_.status1  = status1
mode1_vision_system_.status2  = status2
mode1_vision_system_.status3  = status3
mode1_vision_system_.status4  = status4
mode1_vision_system_.status5  = status5
mode1_vision_system_.status6  = status6
mode1_vision_system_.status7  = status7
mode1_vision_system_.status8  = status8
```

```
mode1_vision_system_.status9  = status9
mode1_vision_system_.status10  = status10

mode1_vision_system_.image1 = image1
mode1_vision_system_.image2 = image2
mode1_vision_system_.image3 = image3
mode1_vision_system_.image4 = image4
mode1_vision_system_.image5 = image5
mode1_vision_system_.image6 = image6
mode1_vision_system_.image7 = image7
mode1_vision_system_.image8 = image8
mode1_vision_system_.image9 = image9
mode1_vision_system_.image10 = image10

mode1_vision_system_.save()


mode1_visionSystem_obj_afterInsertion = mode1_vision_system.objects.latest('date')
mode_selected_obj_first = mode_selected.objects.first()
mode_selected_obj = mode_selected.objects.latest('date')

date1 = mode_selected_obj_first.date
date2 = mode1_visionSystem_obj_afterInsertion.date

def numOfDays(date1, date2):
    return (date2-date1).days

mode_selected_.daysCounter = numOfDays(date1, date2) + 1
mode_selected_.grid = mode_selected_obj.grid
mode_selected_.rows = mode_selected_obj.rows
mode_selected_.columns = mode_selected_obj.columns
mode_selected_.modeNumber = mode_selected_obj.modeNumber
mode_selected_.image = '../assets/gardenPics/' + getTime + '.jpg'
mode_selected_.save()

json = {
'image_json': str(mode1_vision_system_.image),
'cameraDateJSON': str(datetime.now().strftime('%b. %d, %Y, %-I:%M %p')),
'daysCounter_json' : str(numOfDays(date1, date2) + 1),

'plant1_json': mode1_vision_system_.plant1,
'plant2_json': mode1_vision_system_.plant2,
'plant3_json': mode1_vision_system_.plant3,
'plant4_json': mode1_vision_system_.plant4,
'plant5_json': mode1_vision_system_.plant5,
'plant6_json': mode1_vision_system_.plant6,
'plant7_json': mode1_vision_system_.plant7,
'plant8_json': mode1_vision_system_.plant8,
'plant9_json': mode1_vision_system_.plant9,
'plant10_json': mode1_vision_system_.plant10,

'status1_json': mode1_vision_system_.status1,
'status2_json': mode1_vision_system_.status2,
'status3_json': mode1_vision_system_.status3,
'status4_json': mode1_vision_system_.status4,
'status5_json': mode1_vision_system_.status5,
'status6_json': mode1_vision_system_.status6,
```

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        'status7_json': mode1_vision_system_.status7,
        'status8_json': mode1_vision_system_.status8,
        'status9_json': mode1_vision_system_.status9,
        'status10_json': mode1_vision_system_.status10,

        'image1_json': mode1_vision_system_.image1,
        'image2_json': mode1_vision_system_.image2,
        'image3_json': mode1_vision_system_.image3,
        'image4_json': mode1_vision_system_.image4,
        'image5_json': mode1_vision_system_.image5,
        'image6_json': mode1_vision_system_.image6,
        'image7_json': mode1_vision_system_.image7,
        'image8_json': mode1_vision_system_.image8,
        'image9_json': mode1_vision_system_.image9,
        'image10_json': mode1_vision_system_.image10,
        }

        return JsonResponse(json)

    if(mode_selected_obj.modeNumber == 2):
        print(" ")
        print("~[ Mode 2 ] Vision System Starting~")
        print(" ")
        print(" ")

    if(mode_selected_obj.modeNumber == 3):
        print(" ")
        print("~[ Mode 3 ] Vision System Starting~")
        print(" ")
        print(" ")

    if(mode_selected_obj.modeNumber == 4):
        print(" ")
        print("~[ Mode 4 ] Vision System Starting~")
        print(" ")
        print(" ")

if response.POST.get('action') == 'onMode1':

    print(" ")
    print("~Mode 1 Activated~")
    print(" ")

    mode_selected_.grid = mode1_obj_global.grid
    mode_selected_.rows = mode1_obj_global.rows
    mode_selected_.columns = mode1_obj_global.columns
    mode_selected_.modeNumber = mode1_obj_global.modeNumber
    mode_selected_.image = '../assets/background/rpiBG.gif'
    mode_selected_.save()

    mode_selected_obj = mode_selected.objects.latest('date')

    json = {
    'image_json': mode_selected_obj.image,
    'grid_json': mode_selected_obj.grid,
    'mode_json': mode_selected_obj.modeNumber,
    }
```

```
    return JsonResponse(json)

if response.POST.get('action') == 'onMode2':

    print(" ")
    print("~Mode 2 Activated~")
    print(" ")

    mode_selected_.grid = mode2_obj_global.grid
    mode_selected_.rows = mode2_obj_global.rows
    mode_selected_.columns = mode2_obj_global.columns
    mode_selected_.modeNumber = mode2_obj_global.modeNumber
    mode_selected_.image = '../assets/background/rpiBG.gif'
    mode_selected_.save()

    mode_selected_obj = mode_selected.objects.latest('date')

    json = {
    'image_json': mode_selected_obj.image,
    'grid_json': mode_selected_obj.grid,
    'mode_json': mode_selected_obj.modeNumber,
    }

    return JsonResponse(json)

if response.POST.get('action') == 'onMode3':

    print(" ")
    print("~Mode 3 Activated~")
    print(" ")

    mode_selected_.grid = mode3_obj_global.grid
    mode_selected_.rows = mode3_obj_global.rows
    mode_selected_.columns = mode3_obj_global.columns
    mode_selected_.modeNumber = mode3_obj_global.modeNumber
    mode_selected_.image = '../assets/background/rpiBG.gif'
    mode_selected_.save()

    mode_selected_obj = mode_selected.objects.latest('date')

    json = {
    'image_json': mode_selected_obj.image,
    'grid_json': mode_selected_obj.grid,
    'mode_json': mode_selected_obj.modeNumber,
    }

    return JsonResponse(json)

if response.POST.get('action') == 'onMode4':

    print(" ")
    print("~Mode 4 Activated~")
    print(" ")

    mode_selected_.grid = mode4_obj_global.grid
    mode_selected_.rows = mode4_obj_global.rows
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    mode_selected_.columns = mode4_obj_global.columns
    mode_selected_.modeNumber = mode4_obj_global.modeNumber
    mode_selected_.image = '../assets/background/rpiBG.gif'
    mode_selected_.save()

    mode_selected_obj = mode_selected.objects.latest('date')

    json = {
    'image_json': mode_selected_obj.image,
    'grid_json': mode_selected_obj.grid,
    'mode_json': mode_selected_obj.modeNumber,
    }

    return JsonResponse(json)

if response.POST.get('action') == 'onFan':

    print(" ")
    print("~Air Circulation System Activated~")
    print(" ")

    devices_.fansStatus = 'On'
    devices_.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()


if response.POST.get('action') == 'offFan':

    print(" ")
    print("~Air Circulation System Deactivated~")
    print(" ")

    devices_.fansStatus = 'Off'
    devices_.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()

if response.POST.get('action') == 'onWhiteLed':

    print(" ")
    print("~White LED Activated~")
    print(" ")

    devices_.fansStatus = devices_obj_global.fansStatus
    devices_.whiteLedStatus = 'On'
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()
```

```
if response.POST.get('action') == 'offWhiteLed':

    print(" ")
    print("~White LED Deactivated~")
    print(" ")

    devices_.fansStatus = devices_obj_global.fansStatus
    devices_.whiteLedStatus = 'Off'
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()

if response.POST.get('action') == 'onRgbLed':

    print(" ")
    print("~Grow Lights Activated~")
    print(" ")

    devices_.fansStatus = devices_obj_global.fansStatus
    devices_.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = 'On'
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()


if response.POST.get('action') == 'offRgbLed':

    print(" ")
    print("~Grow Lights Deactivated~")
    print(" ")

    devices_.fansStatus = devices_obj_global.fansStatus
    devices_.whiteLedStatus =  devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = 'Off'
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()

if response.POST.get('action') == 'onWater':

    sensors_water = sensors.objects.latest('date')

    if (sensors_water.moisture < 50):
        print(" ")
        print("---------- Automatic Irrigation System Activated ----------")
        print("Time: " + datetime.now().strftime('%Y-%m-%d %H:%M:%S'))
        print("Soil Moisture: " + str(sensors_water.moisture))

        devices_.fansStatus = devices_obj_global.fansStatus
        devices_.whiteLedStatus = devices_obj_global.whiteLedStatus
        devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
```

```
        devices_.calibrationStatus = devices_obj_global.calibrationStatus
        devices_.waterStatus = 'On'
        devices_.seedStatus = devices_obj_global.seedStatus
        devices_.save()

        sleep(1)

        print("Trigger Deactivated...")
        print(" ")
        print(" ")

        devices_2.fansStatus = devices_obj_global.fansStatus
        devices_2.whiteLedStatus = devices_obj_global.whiteLedStatus
        devices_2.rgbLedStatus = devices_obj_global.rgbLedStatus
        devices_2.calibrationStatus = devices_obj_global.calibrationStatus
        devices_2.waterStatus = 'Off'
        devices_2.seedStatus = devices_obj_global.seedStatus
        devices_2.save()

    else:
        print(" ")
        print("Soil Moisture: " + str(sensors_water.moisture))
        print("~Automatic Irrigation System Can't Activate~")
        print(" ")

if response.POST.get('action') == 'onCalibration':

    print(" ")
    print("~Calibration Activated~")
    print(" ")

    devices_.fansStatus = devices_obj_global.fansStatus
    devices_.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_.calibrationStatus = 'On'
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = devices_obj_global.seedStatus
    devices_.save()

    sleep(1)

    print(" ")
    print("~Calibration Deactivated~")
    print(" ")

    devices_2.fansStatus = devices_obj_global.fansStatus
    devices_2.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_2.calibrationStatus = 'Off'
    devices_2.waterStatus = devices_obj_global.waterStatus
    devices_2.seedStatus = devices_obj_global.seedStatus
    devices_2.save()


if response.POST.get('action') == 'onSeed':

    print(" ")
```

```
    print("~Seeding System Activated~")
    print(" ")

    devices_.fansStatus = devices_obj_global.fansStatus
    devices_.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_.calibrationStatus = devices_obj_global.calibrationStatus
    devices_.waterStatus = devices_obj_global.waterStatus
    devices_.seedStatus = 'On'
    devices_.save()

    sleep(1)

    print(" ")
    print("~Seeding System Deactivated~")
    print(" ")

    devices_2.fansStatus = devices_obj_global.fansStatus
    devices_2.whiteLedStatus = devices_obj_global.whiteLedStatus
    devices_2.rgbLedStatus = devices_obj_global.rgbLedStatus
    devices_2.calibrationStatus = devices_obj_global.calibrationStatus
    devices_2.waterStatus = devices_obj_global.waterStatus
    devices_2.seedStatus = 'Off'
    devices_2.save()

if response.POST.get('action') == 'fullReset':

    print(" ")
    print("~Formatting Database, Turning Off Devices...~")
    print(" ")
    print(" ")
    print(" ")

    mode_selected.objects.all().delete()
    mode_selected_.daysCounter = 1
    mode_selected_.grid = mode1_obj_global.grid
    mode_selected_.rows = mode1_obj_global.rows
    mode_selected_.columns = mode1_obj_global.columns
    mode_selected_.modeNumber = mode1_obj_global.modeNumber
    mode_selected_.image = '../assets/background/rpiBG.gif'
    mode_selected_.save()

    devices.objects.all().delete()
    devices_.calibrationStatus = 'Off'
    devices_.fansStatus = 'Off'
    devices_.whiteLedStatus = 'Off'
    devices_.rgbLedStatus = 'Off'
    devices_.waterStatus = 'Off'
    devices_.seedStatus = 'Off'
    devices_.save()

    sensors.objects.all().delete()
    sensors_.temperature = 24
    sensors_.humidity = 65
    sensors_.moisture = 34
    sensors_.temperatureStatus = "Good"
    sensors_.humidityStatus = "Good"
```

```
sensors_.soilMoistureStatus = "Moist"
sensors_.save()

threshold.objects.all().delete()
threshold_.temperature_low = 20
threshold_.temperature_high = 30
threshold_.humidity_low = 49
threshold_.humidity_high = 81
threshold_.moisture_dry = 30
threshold_.moisture_moist = 70
threshold_.moisture_wet = 71
threshold_.save()

mode1.objects.all().delete()
mode1_.grid = '2x5'
mode1_.rows = 2
mode1_.columns = 5
mode1_.modeNumber = 1
mode1_.save()

mode2.objects.all().delete()
mode2_.grid = '2x4'
mode2_.rows = 2
mode2_.columns = 4
mode2_.modeNumber = 2
mode2_.save()

mode3.objects.all().delete()
mode3_.grid = '3x6'
mode3_.rows = 3
mode3_.columns = 6
mode3_.modeNumber = 3
mode3_.save()

mode4.objects.all().delete()
mode4_.grid = '3x4'
mode4_.rows = 3
mode4_.columns = 4
mode4_.modeNumber = 4
mode4_.save()

mode1_vision_system.objects.all().delete()
mode1_vision_system_.image = '../assets/background/rpiBG.gif'
mode1_vision_system_.plant1 = 0
mode1_vision_system_.plant2 = 0
mode1_vision_system_.plant3 = 0
mode1_vision_system_.plant4 = 0
mode1_vision_system_.plant5 = 0
mode1_vision_system_.plant6 = 0
mode1_vision_system_.plant7 = 0
mode1_vision_system_.plant8 = 0
mode1_vision_system_.plant9 = 0
mode1_vision_system_.plant10 = 0
mode1_vision_system_.status1 = 'Early'
mode1_vision_system_.status2 = 'Early'
mode1_vision_system_.status3 = 'Early'
mode1_vision_system_.status4 = 'Early'
```

```
mode1_vision_system_.status5 = 'Early'
mode1_vision_system_.status6 = 'Early'
mode1_vision_system_.status7 = 'Early'
mode1_vision_system_.status8 = 'Early'
mode1_vision_system_.status9 = 'Early'
mode1_vision_system_.status10 = 'Early'
mode1_vision_system_.image1 = '../assets/icons/seedIcon.png'
mode1_vision_system_.image2 = '../assets/icons/seedIcon.png'
mode1_vision_system_.image3 = '../assets/icons/seedIcon.png'
mode1_vision_system_.image4 = '../assets/icons/seedIcon.png'
mode1_vision_system_.image5 = '../assets/icons/seedIcon.png'
mode1_vision_system_.image6 = '../assets/icons/seedIcon.png'
mode1_vision_system_.image7 = '../assets/icons/seedIcon.png'
mode1_vision_system_.image8 = '../assets/icons/seedIcon.png'
mode1_vision_system_.image9 = '../assets/icons/seedIcon.png'
mode1_vision_system_.image10 = '../assets/icons/seedIcon.png'
mode1_vision_system_.save()

mode2_vision_system.objects.all().delete()
mode2_vision_system_.image = '../assets/background/rpiBG.gif'
mode2_vision_system_.plant1 = 0
mode2_vision_system_.plant2 = 0
mode2_vision_system_.plant3 = 0
mode2_vision_system_.plant4 = 0
mode2_vision_system_.plant5 = 0
mode2_vision_system_.plant6 = 0
mode2_vision_system_.plant7 = 0
mode2_vision_system_.plant8 = 8
mode2_vision_system_.plant10 = 0
mode2_vision_system_.status1 = 'Early'
mode2_vision_system_.status2 = 'Early'
mode2_vision_system_.status3 = 'Early'
mode2_vision_system_.status4 = 'Early'
mode2_vision_system_.status5 = 'Early'
mode2_vision_system_.status6 = 'Early'
mode2_vision_system_.status7 = 'Early'
mode2_vision_system_.status8 = 'Early'
mode2_vision_system_.image1 = '../assets/icons/seedIcon.png'
mode2_vision_system_.image2 = '../assets/icons/seedIcon.png'
mode2_vision_system_.image3 = '../assets/icons/seedIcon.png'
mode2_vision_system_.image4 = '../assets/icons/seedIcon.png'
mode2_vision_system_.image5 = '../assets/icons/seedIcon.png'
mode2_vision_system_.image6 = '../assets/icons/seedIcon.png'
mode2_vision_system_.image7 = '../assets/icons/seedIcon.png'
mode2_vision_system_.image8 = '../assets/icons/seedIcon.png'
mode2_vision_system_.save()

mode3_vision_system.objects.all().delete()
mode3_vision_system_.image = '../assets/background/rpiBG.gif'
mode3_vision_system_.plant1 = 0
mode3_vision_system_.plant2 = 0
mode3_vision_system_.plant3 = 0
mode3_vision_system_.plant4 = 0
mode3_vision_system_.plant5 = 0
mode3_vision_system_.plant6 = 0
mode3_vision_system_.plant7 = 0
mode3_vision_system_.plant8 = 0
```

```
mode3_vision_system_.plant9 = 0
mode3_vision_system_.plant10 = 0
mode3_vision_system_.plant11 = 0
mode3_vision_system_.plant12 = 0
mode3_vision_system_.plant13 = 0
mode3_vision_system_.plant14 = 0
mode3_vision_system_.plant15 = 0
mode3_vision_system_.plant16 = 0
mode3_vision_system_.plant17 = 0
mode3_vision_system_.plant18 = 18
mode3_vision_system_.status1 = 'Early'
mode3_vision_system_.status2 = 'Early'
mode3_vision_system_.status3 = 'Early'
mode3_vision_system_.status4 = 'Early'
mode3_vision_system_.status5 = 'Early'
mode3_vision_system_.status6 = 'Early'
mode3_vision_system_.status7 = 'Early'
mode3_vision_system_.status8 = 'Early'
mode3_vision_system_.status9 = 'Early'
mode3_vision_system_.status10 = 'Early'
mode3_vision_system_.status11 = 'Early'
mode3_vision_system_.status12 = 'Early'
mode3_vision_system_.status13 = 'Early'
mode3_vision_system_.status14 = 'Early'
mode3_vision_system_.status15 = 'Early'
mode3_vision_system_.status16 = 'Early'
mode3_vision_system_.status17 = 'Early'
mode3_vision_system_.status18 = 'Early'
mode3_vision_system_.image1 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image2 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image3 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image4 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image5 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image6 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image7 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image8 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image9 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image10 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image11 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image12 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image13 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image14 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image15 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image16 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image17 = '../assets/icons/seedIcon.png'
mode3_vision_system_.image18 = '../assets/icons/seedIcon.png'
mode3_vision_system_.save()

mode4_vision_system.objects.all().delete()
mode4_vision_system_.image = '../assets/background/rpiBG.gif'
mode4_vision_system_.plant1 = 0
mode4_vision_system_.plant2 = 0
mode4_vision_system_.plant3 = 0
mode4_vision_system_.plant4 = 0
mode4_vision_system_.plant5 = 0
mode4_vision_system_.plant6 = 0
mode4_vision_system_.plant7 = 0
```

```
mode4_vision_system_.plant8 = 0
mode4_vision_system_.plant9 = 0
mode4_vision_system_.plant10 = 0
mode4_vision_system_.plant11 = 0
mode4_vision_system_.plant12 = 12
mode4_vision_system_.status1 = 'Early'
mode4_vision_system_.status2 = 'Early'
mode4_vision_system_.status3 = 'Early'
mode4_vision_system_.status4 = 'Early'
mode4_vision_system_.status5 = 'Early'
mode4_vision_system_.status6 = 'Early'
mode4_vision_system_.status7 = 'Early'
mode4_vision_system_.status8 = 'Early'
mode4_vision_system_.status9 = 'Early'
mode4_vision_system_.status10 = 'Early'
mode4_vision_system_.status11 = 'Early'
mode4_vision_system_.status12 = 'Early'
mode4_vision_system_.image1 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image2 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image3 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image4 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image5 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image6 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image7 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image8 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image9 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image10 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image11 = '../assets/icons/seedIcon.png'
mode4_vision_system_.image12 = '../assets/icons/seedIcon.png'
mode4_vision_system_.save()

mode_selected_obj = mode_selected.objects.latest('date')
mode1_visionSystem_obj = mode1_vision_system.objects.latest('date')
sensors_obj = sensors.objects.latest('date')
devices_obj = devices.objects.latest('date')

json = {
'mode_json': mode_selected_obj.modeNumber,
'grid_json': mode_selected_obj.grid,
'startDate_json': str(datetime.now().strftime('%b. %d, %Y, %-I:%M %p')),
'daysCounter_json' : str(mode_selected_obj.daysCounter),

'calibration_json' : devices_obj.calibrationStatus,
'fans_json' : devices_obj.fansStatus,
'whiteLed_json' : devices_obj.whiteLedStatus,
'rgbLed_json' : devices_obj.rgbLedStatus,
'water_json' : devices_obj.waterStatus,
'seeder_json' : devices_obj.seedStatus,

'temperature_json': sensors_obj.temperature,
'humidity_json': sensors_obj.humidity,
'soilMoisture_json': sensors_obj.moisture,
'temperatureStatus_json': sensors_obj.temperatureStatus,
'humidityStatus_json': sensors_obj.humidityStatus,
'soilMoistureStatus_json': sensors_obj.soilMoistureStatus,

'image_json' : str(mode1_visionSystem_obj.image),
```

```
        }

        print("-------------------------- GrowSmart Initializing ----------------------------")
        print("Time: " + datetime.now().strftime('%Y-%m-%d %H:%M:%S'))
        print("Mode: " + str(mode_selected_obj.modeNumber))
        print("Grid: " + mode_selected_obj.grid)
        print(" ")
        print(" ")

        return JsonResponse(json)

    sensors_obj_global = sensors.objects.latest('date')
    mode1_vision_system_obj_global = mode1_vision_system.objects.latest('date')
    mode2_vision_system_obj_global = mode2_vision_system.objects.latest('date')
    mode3_vision_system_obj_global = mode3_vision_system.objects.latest('date')
    mode4_vision_system_obj_global = mode4_vision_system.objects.latest('date')
    mode_selected_obj_global_first = mode_selected.objects.first()
    mode_selected_obj_global_2 = mode_selected.objects.latest('date')

    myObj = {'mode_selected_obj_global_first': mode_selected_obj_global_first, 'mode_selected_obj_global_2':
mode_selected_obj_global_2, 'devices_obj_global': devices_obj_global,
            'sensors_obj_global': sensors_obj_global, 'mode1_vision_system_obj_global': mode1_vision_system_obj_global,
'mode2_vision_system_obj_global': mode2_vision_system_obj_global
            , 'mode3_vision_system_obj_global': mode3_vision_system_obj_global, 'mode4_vision_system_obj_global':
mode4_vision_system_obj_global }

    return render(response, 'main.html', context=myObj)

def databasePage(response):

    devices_obj_global = devices.objects.all()
    sensors_obj_global = sensors.objects.all()
    mode1_vision_system_obj_global = mode1_vision_system.objects.all()
    mode_selected_obj_global = mode_selected.objects.all()
    mode_selected_obj_global_2 = mode_selected.objects.latest('date')

    myObj = {'mode_selected_obj_global': mode_selected_obj_global, 'mode_selected_obj_global_2': mode_selected_obj_global_2,
'devices_obj_global': devices_obj_global,
            'sensors_obj_global': sensors_obj_global, 'mode1_vision_system_obj_global': mode1_vision_system_obj_global}

    return render(response, 'database.html', context=myObj)

def sensorsPage(response):

    threshold_ = threshold()

    if response.POST.get('action') == 'temperature_low_set':

        threshold_obj0 = threshold.objects.latest('date')

        threshold_.temperature_low = response.POST.get('temperature_low_data')
        threshold_.temperature_high = threshold_obj0.temperature_high
        threshold_.humidity_low = threshold_obj0.humidity_low
        threshold_.humidity_high = threshold_obj0.humidity_high
        threshold_.moisture_dry = threshold_obj0.moisture_dry
        threshold_.moisture_wet = threshold_obj0.moisture_wet
        threshold_.save()
```

```
    threshold_obj = threshold.objects.latest('date')

    json = {
    'temperature_low' : threshold_obj.temperature_low
    }
    return JsonResponse(json)

if response.POST.get('action') == 'temperature_high_set':

    threshold_obj0 = threshold.objects.latest('date')

    threshold_.temperature_low = threshold_obj0.temperature_low
    threshold_.temperature_high = response.POST.get('temperature_high_data')
    threshold_.humidity_low = threshold_obj0.humidity_low
    threshold_.humidity_high = threshold_obj0.humidity_high
    threshold_.moisture_dry = threshold_obj0.moisture_dry
    threshold_.moisture_wet = threshold_obj0.moisture_wet
    threshold_.save()

    threshold_obj = threshold.objects.latest('date')

    json = {
    'temperature_high' : threshold_obj.temperature_high
    }
    return JsonResponse(json)

if response.POST.get('action') == 'humidity_low_set':

    threshold_obj0 = threshold.objects.latest('date')

    threshold_.temperature_low = threshold_obj0.temperature_low
    threshold_.temperature_high = threshold_obj0.temperature_high
    threshold_.humidity_low = response.POST.get('humidity_low_data')
    threshold_.humidity_high = threshold_obj0.humidity_high
    threshold_.moisture_dry = threshold_obj0.moisture_dry
    threshold_.moisture_wet = threshold_obj0.moisture_wet
    threshold_.save()

    threshold_obj = threshold.objects.latest('date')

    json = {
    'humidity_low' : threshold_obj.humidity_low
    }
    return JsonResponse(json)

if response.POST.get('action') == 'humidity_high_set':

    threshold_obj0 = threshold.objects.latest('date')

    threshold_.temperature_low = threshold_obj0.temperature_low
    threshold_.temperature_high = threshold_obj0.temperature_high
    threshold_.humidity_low = threshold_obj0.humidity_low
    threshold_.humidity_high = response.POST.get('humidity_high_data')
    threshold_.moisture_dry = threshold_obj0.moisture_dry
    threshold_.moisture_wet = threshold_obj0.moisture_wet
    threshold_.save()
```

```
    threshold_obj = threshold.objects.latest('date')

    json = {
    'humidity_high' : threshold_obj.humidity_high
    }
    return JsonResponse(json)

if response.POST.get('action') == 'moisture_dry_set':

    threshold_obj0 = threshold.objects.latest('date')

    threshold_.temperature_low = threshold_obj0.temperature_low
    threshold_.temperature_high = threshold_obj0.temperature_high
    threshold_.humidity_low = threshold_obj0.humidity_low
    threshold_.humidity_high = threshold_obj0.humidity_high
    threshold_.moisture_dry = response.POST.get('moisture_dry_data')
    threshold_.moisture_wet = threshold_obj0.moisture_wet
    threshold_.save()

    threshold_obj = threshold.objects.latest('date')

    json = {
    'moisture_dry' : threshold_obj.moisture_dry
    }
    return JsonResponse(json)

if response.POST.get('action') == 'moisture_wet_set':

    threshold_obj0 = threshold.objects.latest('date')

    threshold_.temperature_low = threshold_obj0.temperature_low
    threshold_.temperature_high = threshold_obj0.temperature_high
    threshold_.humidity_low = threshold_obj0.humidity_low
    threshold_.humidity_high = threshold_obj0.humidity_high
    threshold_.moisture_dry = threshold_obj0.moisture_dry
    threshold_.moisture_wet = response.POST.get('moisture_wet_data')
    threshold_.save()

    threshold_obj = threshold.objects.latest('date')

    json = {
    'moisture_wet' : threshold_obj.moisture_wet
    }
    return JsonResponse(json)

threshold_obj_global = threshold.objects.latest('date')

myObj = {'threshold_obj_global': threshold_obj_global}

return render(response, 'sensors.html', context=myObj)
```